

Sprite Sheet Generation using a Diffusion Model

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Introduction: Sprite sheets are a fundamental component in 2D game development, which stack a series of frames to form character animations. The manual, frame-by-frame creation of these assets is a demanding task that requires extensive time and artistic skill. As image generative models lack temporal awareness, applying them directly for sprite sheet generation leads to flickering and identity drift across frames. Although video generation models like AnimateAnyone [3] address temporal consistency, they are not tailored for the specific stylistic and structural requirements of game assets. Furthermore, a significant gap exists in generating game characters directly from realistic human portraits, a key feature for enabling user personalisation in games and metaverse applications.

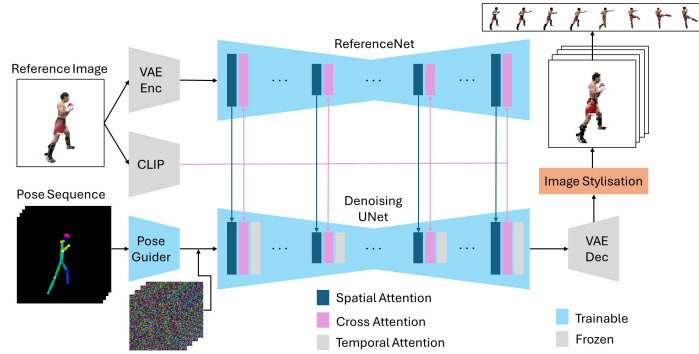


Figure 1: The model architecture

Method: We present a pipeline that transforms a single human portrait into an animated pixel art sprite sheet. Our model architecture, shown in Fig. 1, is adapted from AnimateAnyone [3] and Sprite Sheet Diffusion [2]. It is built upon a pre-trained Stable Diffusion v1.5 model. The key components are: (i) *ReferenceNet* - A U-Net that encodes appearance features from the input portrait; (ii) *Pose Guider* - A lightweight module, inspired by ControlNet, that injects pose sequence information into the main diffusion process; (iii) *Denoising U-Net with Motion Module* - The core generative network, which includes temporal attention layers (Motion Module) to ensure coherence across frames. During generation, features from the ReferenceNet and Pose Guider are passed into the Denoising U-Net via cross-attention and additive mechanisms, guiding the synthesis to match the input identity and target poses. The loss function can be presented as:

$$L = E_{Z_0, I_{ref}, P, \epsilon \sim \mathcal{N}(0,1), t} \left[w(t) \cdot \|\mathbf{v}_{target} - \epsilon_{\theta}(Z_t, t, I_{ref}, P)\|^2 \right] \quad (1)$$

where I_{ref} : Reference portrait image; P : Sequence of pose skeletons $p_1, \dots, p_N$; E : VAE Encoder, so $Z_0 = E(I_{ref})$; Z_t : Noisy latent sequence at timestep t ; $\mathbf{v}_{target}$: The target velocity for v-prediction; $w(t)$: SNR-based loss weight at timestep t ; ϵ_{θ} : The complete Denoising U-Net, which internally contains the ReferenceNet, Pose Guider, and Motion Module.

To fine-tune our model, we created a paired dataset from publicly available videos of human motion. The dataset contains 845 frames across 94 distinct actions (e.g., walking, jumping, attacking). The videos went through preprocessing steps like manual keyframe extraction, background removal with U2-Net [4] and pose skeleton extraction with OpenPose [1] in COCO-18 keypoint format. To simulate the pixel art style from human portraits, an image stylization technique, "pixelation", is implemented. This involves downsampling the image, upsampling using nearest neighbour interpolation to create the blocky effect.

Results and Evaluation: The main focus of our research is to determine the optimal workflow for combining motion generation and stylisation. We designed an ablation study to compare two approaches: (i) *Approach A* - The model generates a realistic human portrait animation based on the pose skeletons. The programmatic pixelation filter is applied as a final post-processing step; (ii) *Approach B* - The input portrait is first pixelated. The model is then trained to generate an animated sequence directly in pixel art style.

We fine-tuned a separate model for each approach on our custom dataset to compare their performance. We evaluated our two fine-tuned models against the original, non-fine tuned Sprite Sheet Diffusion model [2], which serves as our baseline. All models were trained for 30,000 steps on an NVIDIA L40S GPU.

Table 1: Overall average evaluation metric results.

Model	SSIM $\uparrow$	PSNR $\uparrow$	LPIPS $\downarrow$	SC $\uparrow$
Sprite Sheet Diffusion [2]	0.6544	15.7420	0.2103	0.8547
Approach A	0.7450	24.8354	0.0366	0.8716
Approach B	0.7414	22.5298	0.0642	0.8167

As shown in Table 1, the results show that Approach A is the superior method. It achieves the best scores on all four metrics, including a markedly lower (better) LPIPS score of 0.0366 and the highest Subject Consistency of 0.8716. This indicates that generating motion on the high-fidelity portrait allows the model to better preserve identity and create more coherent animations. Applying stylisation as a deterministic final step retains this quality. In contrast, Approach B, which works with a lower-information pixelated input, struggles more with consistency, as reflected in its lower SC score.



Figure 2: Output comparison for a "Combo Attack" motion.

To test generalisation, we used the models on unseen personal portraits. As shown in Fig. 3, the fine-tuned model (Approach A) successfully transfers the identity from the new portrait into the animated character. This demonstrates its practical utility for creating personalised avatars.



Figure 3: Example of generalisation to an unseen personal portrait.

Conclusion: This research presents a validated, end-to-end pipeline for automating the creation of personalised 2D sprite sheets character animations, addressing a key bottleneck in game development.

References:

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